

L13: Topology Inference Problems

Let us now move from network sampling to a different problem: **topology inference**. *How can we estimate the topology of a network from incomplete information about its nodes and edges?*

The problem of topology inference has several variations, described below. The following visualization helps to illustrate each variation. The top-left figure shows the actual (complete) network, which consists of five nodes and five edges (shown in solid dark blue) -- the dotted dark blue lines represent node pairs that are NOT connected with an edge. The topology inference problems we will cover in the following pages are:

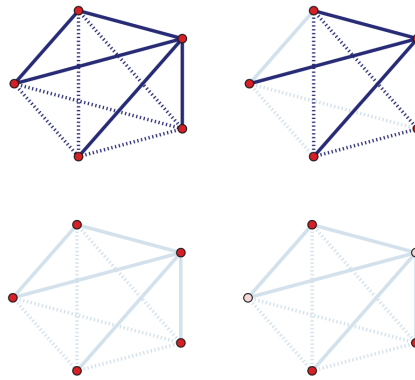


Image source: Kolaczyk, Eric D. *Statistical Analysis of Network Data: Methods and Models* (2009) Springer Science+Business Media LLC.

a) Link prediction: As shown in the top-right figure, in some cases we know that certain node pairs are connected with an edge (solid dark blue) or that they are NOT connected with an edge (dotted dark blue) -- but we do not know what happens with the remaining node pairs. Are they connected or not? In the top-right figure those node pairs are shown with solid light blue and dotted light blue, respectively. We have already mentioned the link prediction problem in Lesson-12, in the context of the Hierarchical Random Graph (HRG) model. Here, we will see how to solve this problem even if we do not model the network using HRG.

b) Association networks: As shown in the bottom-left figure, in some cases we know the set of nodes but we do not have any information about the edges. Instead, we have some data about various characteristics of the nodes, such as their temporal activity or their attributes. Imagine, for instance, that we know all the characteristics, hobbies, interests, etc, of the students in a class, and we try to infer who is friend with whom. We can use the node attributes to identify pairs of nodes that are highly similar according to a given metric. Node pairs that are highly similar are then assumed to be connected.

c) Network tomography: As shown in the bottom-right figure, in some cases we know some nodes (shown with red) but we do not know about the existence of some other nodes (shown in pink) -- and we may not know the edges either. This is clearly the hardest topology inference problem in networks but sufficient progress has been made in solving it as long as we can make some "on-demand path measurements" from the nodes that we know of. Additionally, the problem of network tomography is significantly simpler if we can make the assumption that the underlying network has a tree topology (this is not the case in the given example).